AquaSave: Smarter Water, Sustainable Cities

Tackling critical urban water wastage, AquaSave is a pioneering sustainable project. It intelligently integrates advanced RO reject water recovery, innovative 3D-printed smart storage technology, and precise, automated soil moisture sensor irrigation systems to create more efficient and sustainable cities.

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The Critical Problem: RO Water Wastage

The Shocking Reality

Conventional reverse osmosis (RO) water purifiers reject **75%–80%** of input water during the filtration process. An average family wastes approximately **15,000 litres of water annually**—clean, usable water discharged directly into drainage systems.

This represents not merely an environmental concern, but a significant economic burden on households and communities facing increasing water scarcity.

Why This Matters

- Contributes directly to urban water scarcity in growing cities
- Represents avoidable wastage of perfectly usable greywater
- Unnecessarily increases monthly household water bills
- Contradicts global sustainability targets

The Hidden Opportunity in Rejected Water

Cleaning Utensils

Perfectly safe for washing dishes and cookware

Floor Mopping

Ideal for daily household floor cleaning

Washing Clothes

Suitable for laundry and textile cleaning

Watering Plants

Excellent for indoor and outdoor plant irrigation

Despite these valuable applications, **no affordable system currently captures and repurposes this rejected water** for household use, representing a significant market gap and environmental opportunity.

Introducing AquaSave

Smart RO Reject Water Recycling System

01

Collects RO Waste Water

Automatically captures rejected water during filtration

02

Stores in 3D-Printed Canister

Custom-designed, space-efficient storage solution

03

Redirects for Household Uses

Makes water available for non-potable applications

04

Automates Plant Irrigation

Electronic sensors trigger watering when needed

AquaSave represents a **low-cost, modular system** that transforms waste into a valuable household resource through intelligent automation and sustainable design principles.

System Components & Technical Architecture

1

3D-Printed Water Storage Canister

- Custom-designed, space-saving form factor
- Durable, lightweight PLA or PETG material
- Integrated spout with pipe connectors
- Printable on standard FDM 3D printers
- 15-litre capacity optimised for household use

2

RO Waste Water Capture System

- Quick-connect pipe fittings for easy installation
- Flow-control valve for regulation
- Optional filter mesh for debris removal
- Compatible with standard RO outlet connections

3

Automatic Plant-Irrigation Module

- Calibrated soil moisture sensor
- Arduino/ESP-based microcontroller
- 5V submersible water pump
- Auto-trigger irrigation based on soil conditions
- Low power consumption design

How AquaSave Works

Water Redirection

RO reject water is automatically redirected to the AquaSave canister through a seamless connection system

Level Monitoring

Integrated sensors continuously track water levels inside the canister, preventing overflow

Soil Analysis

Programmable circuit monitors soil moisture levels in real-time using embedded sensors

Automated Watering

When soil becomes dry, the mini pump activates automatically, delivering precise water quantities

Multi-Purpose Use

Remaining stored water remains available for household cleaning, mopping, and other non-potable uses

Target Market & Customer Segments

Primary Consumers

Residential Homes with RO Systems

Middle-class families seeking water conservation solutions

Small Offices & Commercial Spaces

Businesses aiming to reduce operational costs and meet sustainability targets

Apartment Complexes

Multi-family dwellings with centralised water management needs

Educational Institutions

Schools and small institutions promoting environmental responsibility

Why They'll Invest in AquaSave

- Demonstrable water savings
- Reduced monthly consumption and bills
- Simple plug-and-play installation
- Zero ongoing maintenance requirements
- Alignment with personal sustainability goals

Our Unique Selling Proposition

Retrofittable Solution

Low-cost design works with existing RO systems—no expensive replacements needed

Zero Plumbing Changes

Quick-connect fittings mean installation in minutes without professional help

3D Printed Flexibility

Customisable designs, easily repairable components, and local manufacturing potential

Smart Automation

Automated irrigation eliminates manual watering whilst optimising water usage

Environmental Impact

Contributes directly to water conservation and circular economy principles

Financial Breakdown & Pricing Strategy

Per Unit Production Cost Analysis

Component	Specification	Cost
15L Storage Tank (3D-Printed)	Durable PLA/PETG canister	₹850
Pipes, Fittings, Valves	Leak-proof quick connectors	₹200
Low-Power DC Pump	5V irrigation automation	₹380
Soil Moisture Sensor + Circuit	Arduino-based control module	₹150
Assembly, Labour, Packaging	Finishing and quality control	₹150
Total Cost Price (COP)	—	**₹1,730**
Suggested Selling Price (SSP)	Competitive market pricing	**₹2,200**
Profit Margin per Unit	SSP – COP	**₹470 (27%)**

This pricing strategy ensures **affordability for mass markets** whilst maintaining healthy margins for sustainable business growth and reinvestment in product development.

Investment Proposal

₹4,00,000

For 20% Equity

Strategic Fund Allocation

1

Production Scaling

Mass-printing of 3D canister units using advanced manufacturing

2

Initial Inventory

First 100 complete production kits for market testing

3

Brand Development

Professional packaging, branding materials, and product documentation

4

Distribution Network

School-level and community distribution channels

5

Marketing & Testing

Customer acquisition campaigns and pilot programme feedback

Environmental Impact & Vision

14.6K

Litres Saved Annually

Per household using AquaSave system

75%

Water Recovery Rate

From previously wasted RO reject water

100

Initial Production Units

First wave deployment across pilot markets

Long-Term Vision for Circular Water Economy

AquaSave represents more than a product—it's a movement towards comprehensive greywater management and climate responsibility. By empowering students to engage with **3D design, electronics, sensor automation, market research, and sustainable engineering**, this capstone project demonstrates how innovation can address real-world environmental challenges.

> **AquaSave turns wasted water into a valuable resource—every single day.**

This project embodies the intersection of innovation, sustainability, and entrepreneurship, preparing the next generation to build solutions for a water-secure future.